Affan Ahmad Basra

Data Engineer | Python, SQL, PostgreSQL, AWS | ML and LLM Pipelines

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Summary

Data engineer with 1.5 years of experience building production data pipelines and ML/LLM systems. Built and ran ETL pipelines for two US sports data companies, managed a production AWS RDS PostgreSQL database, and migrated a startup's database with no data lost. Works directly with US clients in English. Second focus on LLM pipelines and ML deployment.

Technical Skills

Data Engineering	Python (Polars, pandas, NumPy), SQL, PostgreSQL, ETL/ELT pipelines, schema validation, data modeling
Cloud & DevOps	AWS (RDS), Docker, Docker Compose, GitHub Actions CI/CD, Git, pytest
Backend & Streaming	Apache Kafka, REST APIs, Django, SQLAlchemy, Alembic, Streamlit
ML & LLM	LangChain, LangGraph, RAG, PyTorch, Hugging Face, scikit-learn, model deployment and monitoring
Currently Learning	Spark, Airflow, dbt

Education

BS Data Science, National University of Sciences and Technology (NUST) **Languages:** English (fluent), Urdu (native)
Certifications (DeepLearning.AI, Coursera): Machine Learning in Production | Data Engineering Specialization | Deep Learning Specialization

Experience

ByteSquad Jul 2025 – Aug 2026
Data Engineer (Contract), for US clients StatRankings and DFS Army *Remote*

- Built and ran ETL pipelines delivering NFL and NBA stats on daily and weekly schedules, 10K+ rows a week feeding multiple downstream platforms.
- Wrote the pipeline code and source tables in Python (Polars) and loaded the data into PostgreSQL on AWS RDS.
- Enforced data quality with schema checks on every load (NOT NULL, type, unique and CHECK constraints), so bad loads failed before reaching client tables; set up Slack and email failure alerts with timestamped error logs.
- Traced a wrong-stat incident from the front end back to undocumented empty fields in a source API, and rebuilt the stat from a second API.
- Managed the production AWS RDS database for DFS Army (tables, indexes, backups, monitoring) and delivered both projects end to end, working directly with US clients.

Aawaz AI Jun 2025 – Aug 2025
ML / Data Engineer *Islamabad, Pakistan*

- Built a ReAct agent in LangChain with tool calling and REST APIs for competitor research, reaching 90% task accuracy.
- Built a multi-step LLM workflow with Gemini and Django for marketing strategy generation, cutting manual effort by 75%.
- Added RAG retrieval over document stores and conversation caching.

DLL-TUKL Jun 2024 – Aug 2024
ML / AI Intern *Islamabad, Pakistan*

- Built a deep learning text-to-speech pipeline reaching 97% speech accuracy, reducing manual verification by 40%.
- Tuned a conversational LLM for inference speed, cutting user wait times by 35%.

Projects

PipelineGuard: Streaming PII Firewall | *Python, Kafka, PostgreSQL, GLiNER, Docker* GitHub | Live demo

- Built a Kafka stream processor that redacts Pakistani PII: rules for CNIC, IBAN (mod-97 check), phone and email, plus a GLiNER encoder for names and addresses.
- Wrote a PostgreSQL audit trail that stores entity type, span and confidence, never the value. At-least-once delivery with idempotent writes: no loss or duplication across a 2,500-message crash-and-replay test.
- Compared 4 encoder models and reached 96.0% address coverage on 2,786 real OpenStreetMap addresses. Micro-batching raised throughput about 180x, to 1,267 records/s.
- Cut model memory from 1,780 MB to 845 MB with bf16 at unchanged coverage, and deployed a live demo on Streamlit Cloud. 509 tests run in GitHub Actions CI.

Database Migration, Supabase to Neon | *PostgreSQL 17, SQLAlchemy, Alembic, Docker* Client project

- Migrated a startup's production database with pg_dump and pg_restore. Cleaned stale data first, verified all 11 tables by row counts and checksums, and cut over at night with a brief write pause and no data lost. Rollback in about 2 minutes.